Revulagadda Rahul

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	6.97
Lakshya School	Class XII (CBSE)	2023	88.60%
Aditya High school	Class X (SSC)	2021	92.83%

SKILLS

Web Development: HTML, CSS, Express.js ,React.js ,Node.js

Languages: C++, C , Verilog, Python

Developer Tools: Git, Linux, Latex, Docker

Computational Tools: Solid edge, Matlab, Ltspice, Ngspice, Vivado

Databases: MySQL, MongoDB

Competitive Programming: Solved 250+ problems on Leetcode.

PROJECTS

Cartify - eCommerce Web App

Tech Stack: *MongoDB, Express.js, React.js, Node.js, JWT, Stripe*

- Built a full-stack eCommerce platform with **7 React pages**, enabling product browsing, cart management, user authentication, checkout, and order tracking, while delivering a seamless end-to-end shopping experience across the application.
- Developed **12+ RESTful APIs** using **Express.js, MongoDB, and Mongoose** for user authentication, product CRUD operations, cart management, and order processing, following a structured and modular backend architecture.
- Implemented **JWT-based authentication with bcrypt password hashing** and protected user/admin routes, while organizing backend data management through **3 Mongoose models: User, Product, and Order** for efficient application functionality.
- Built a responsive React frontend using **reusable components and Axios API integration**, implemented persistent shopping-cart functionality, and integrated **Stripe payment processing** to support a complete and reliable checkout workflow.

Parallel Fast Fourier Transform using Multithreading

Tech Stack: C++, C++17, POSIX Threads, Parallel Computing, FFT

- Implemented **6 FFT variants**, including recursive and iterative **Cooley-Tukey and Bailey FFT algorithms**, covering both sequential and parallel implementations to study different approaches to efficient frequency-domain computation.
- Developed **3 multithreaded FFT implementations using POSIX Threads**, distributing computational workloads across multiple threads and optimizing thread-level workload allocation for efficient parallel execution on **power-of-2 input sizes**.
- Benchmarked sequential and parallel FFT implementations across different **input sizes and thread counts**, measuring execution time and analyzing the performance impact of increasing thread-level parallelism on overall computation.
- Performed comparative analysis of the implemented FFT algorithms using **execution-time measurements and experimental results**, evaluating computational efficiency and documenting observations across different configurations.

Active Noise Cancellation (ANC)

Tech Stack*: MATLAB, RLS Adaptive Filtering, Digital Notch Filtering*

- Designed an **adaptive noise cancellation system** using DSP techniques and adaptive filtering for effective noise suppression.
- Implemented **RLS adaptive filtering** for full-mode noise suppression with improved convergence and efficiency.
- Designed **digital notch filters** to selectively suppress tonal noise while preserving the desired audio signal.
- Developed **Full and Partial Suppression modes,** evaluated performance by noise reduction and audio quality analysis.

Numerical Methods

Tech stack*: Python, NumPy, SciPy, Jupyter Notebook, Streamlit*

- Implemented **Euler, Runge-Kutta, integration, differentiation, LU decomposition, root-finding, and interpolation.**
- Optimized numerical algorithms for **stability, accuracy, computational efficiency, and correctness** across problems.
- Built **interactive Streamlit interfaces** enabling users to input problems, execute algorithms, and visualize results.
- Integrated **NumPy and SciPy** for numerical validation and organized implementations into reusable computational modules.

RELEVANT COURSES

Computer Science: Introduction to Programming, DBMS, Data Structures and Applications

AI :Introduction to AI , Machine Learning, Probability and Random variables

Mathematics: Calculus-1, Calculus-2, Transform Techniques,Matrix Theory,Complex Variables, Differential Equations,

Misc: Energy Management,Digital Fabrication

EXTRACURRICULAR

- Worked as a volunteer in the events conducted by the NSS organisation.
- worked as a member of Canteen Monitoring Committee .
- regularly plays volley ball and participate in the sports event.

POSITIONS OF RESPONSIBILITY

Mentor - Sunshine, Mental Well-being Club

- Mentored 40+ juniors through regular interactions, organizing engaging activities and supporting them in navigating personal and academic challenges.

Event Coordinator - Abhyant, IIT Hyderabad's Inaugural Sports Fest

- Managed on-ground operations and end-to-end event execution, coordinating across multiple domains to ensure seamless event conduction.